

Using Computer Vision for Local Waste Detection,
Classification, and Mapping: a novel approach to optimizing
model accuracy with Instance Segmentation

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MSEF 2026: Submitted March 4, 2026

1 Abstract

Rampant litter densities are a global issue as industrialization has increased consumer activity, resulting in urban regions with high volumes of trash, with little being known about the specific distribution of these items. Due to the issues of ecosystem health, microplastic emissions, and water pollution, increasing efforts have been initiated to help track and map litter products within regions with the goals of removing litter products or identifying which regions require the most aid. One solution is crowdmapping, a process where community members mark specific regions corresponding to identifying litter and add such instances to a growing database. Applications of crowdmapping are enhanced through the automation of detection, a practice that this project aims to facilitate by engineering a set of Computer Vision models for identifying a wide range of litter products. This design enables the complete identification and localization of 7 litter object varieties within images, with full inference speed across all models combined under 1 second. The developed softwares also show promising accuracy over existing frameworks: as demonstrated through most model F1 scores in the range 0.8 to 0.9 and mAP50 values from 0.8 to 0.95 across models compared to the current benchmarks of roughly 0.7 F1 scores and 0.65 mAP50 currently used in similar tasks. As a proposed application of this software, the trained models are being implemented into an autonomous litter crowdmapping software capable of user upload and litter identification based on uploaded images.

2 Introduction

2.1 A Note Before Starting

The motivation for this project came from participation in a community-wide cleanup initiative taking place in late 2025. During this event, volunteers relied primarily on subjective assessments from organizers to determine where litter accumulation was likely to be highest. These recommendations were based on general observation rather than quantitative data, highlighting a broader limitation in how environmental cleanup efforts identify and prioritize high-density litter areas. This observation prompted consideration of whether a more data-driven approach could improve efficiency. In particular, crowdmapping appeared as a desirable approach which could be drastically enhanced with the use of AI for automation, prompting this project to develop such needed systems.

2.2 Project Summary

This engineering project centers around the development of a set of Computer Vision (CV)—a field of Machine Learning and therefore also AI—models (meaning trained scripts used for detecting objects in an input image) each capable of accurately detecting a single object class in visual data. Using the YOLOv11 framework, this project expands upon current measures for detecting litter based on visual survey data and establishes a highly reliable detection system based upon the annotated images of a variety of publicly available image datasets. Then, using another set of images for testing, the trained models are compared against the current benchmarks for the selected object classes to evaluate their overall performance, progress, and use for real-world applications.

The branch of CV used throughout this research is called Instance Segmentation, which, when given an image or frame of a video, produces both bounding boxes (rectangles) around objects of a certain class in the image as well as creates pixel representations for the exact boundaries of any detected objects. While many current CV applications use the standard Object Detection in place of this separate CV framework, capable only of creating bounding boxes, this designed system aims to experiment with the proposed increased precision of Instance Segmentation models to improve application performance of the models.

As a more general issue, litter pollution is one of the most environmentally important factors that determines the health of ecosystems. As a result, many initiatives hope to determine litter product densities to aid in supporting these environments and determining which areas require the most aid.

However, identifying and classifying litter product densities over large regions can be a tedious task when relying completely on manual efforts to analyze visual data in environments. Environmental initiatives either benefit from or require accurate data for waste densities to improve their ability to remove litter objects or analyze patterns in these densities. This is an issue given the lack of reliable ways of automatically locating and mapping high densities within certain urban environments. A

potential solution has been the use of CV to automate the detection of waste products given more easily obtainable visual data of urban regions. Although current innovations in applicable CV systems have the potential for this task, the issue remains of improving upon the accuracy of systems while maintaining real-time model speeds. This is the objective of this design project along with building a real-time mapping system based on CV models for faster detection and labeling.

2.3 Role and use of Computer Vision

Computer Vision refers to the field of study focusing on converting visual data into a processable product for computers. Some of the most common subfields are Image Classification and Object Detection respectively. A less commonly used subfield is Instance Segmentation. The following figure displays the output of Object Detection and Instance Segmentation models when given visual data.



Figure 1: Examples of Instance Segmentation (on right) and Object Detection (on left)

Object Detection models draw bounding boxes around the objects in an image perceived as the desired object or set of objects that the model is trained to detect. It is unable to precisely draw pixel-masks (these are pixel representations of every pixel in the model prediction that is assumed to be the object class) around such objects, this being the improvement that Instance Segmentation offers over its counterpart. Both systems have similar objectives but differ in their outputs.

The YOLOv11 architecture is a very recent CV framework that employs a CNN based architecture for many CV subtasks. As such, due to its reputation for real-world applications based upon high inference speed (defined in section 5.2), this architecture is used as the basis for training these models. This implies that although the basic fundamental detection strategies are based upon this framework, the optimization, hyperparameterization, model training and dataset collection are done using instead of relying on the framework.

While Object Detection relies on rectangular bounding boxes, these boxes inevitably include "background noise" (e.g., pavement, grass, or shadows) that the models associate with the target class. In contrast, Instance Segmentation utilizes precise pixel-level masks that isolate the object's unique features from its environment. By supervising the model with these masks, the model training can more heavily penalize "false associations" with background pixels. This is the basis for our hypothesis: by reducing spatial noise and focusing the CNN on the exact geometric boundaries of irregular litter (like crumpled bags or straws), the model will achieve higher accuracy as a result.

3 Overview of Literature Review and Comparison

Through analyzing current approximate studies on litter data, the following have been recognized as the most prevalent litter products most suited for detection using CV and, as such, models will be developed for these categories:

1. Plastic Bottles
2. Drink Cans
3. Food Containers and Wrappers
4. Plastic Bags
5. Plastic Cups
6. Plastic Films
7. Drinking Straws

To test model accuracy (see section 5.3 for more information on how this comparison will be done), we implement comparisons across a number of prior software benchmarks within the studied object classes to compare the usability of our developed models for such systems. The following table contains the previous benchmark work within each of these studied object classes:

Table 1: Prior Work

Object Class	Source 1	Source 2	Source 3
Plastic Bottles	(Heller, A 2025)	(Jaikumar, P 2022)	(Dipo, M 2025)
Drink Cans	(Mandhana, N 2026)	(Akkoyun, D 2024)	
Plastic Bags		(Akkoyun, D 2024)	
Food Containers and Wrappers			(Shad, I 2025)
Plastic Cups		(Akkoyun, D 2024)	
Plastic Films	(Mandhana, N 2026)	(Akkoyun, D 2024)	
Drinking Straws		(Akkoyun, D 2024)	

These works were selected prior to training the models due to their high accuracy metrics and status as benchmarks for these object classes. The object classes of plastic cups, plastic films and drinking straws were difficult to find widely accessible prior work on due to their uniqueness among object classes and lack of many prior datasets within these classes. However, as their metrics are still at a benchmark level, these models are sufficient for a high-level comparison of accuracy metrics in showing the progress of developed systems from this design process.

3.1 Generalization about the State of the Field

After viewing the current state of many predeveloped models and frameworks for these applications, the following two observations were most clear:

1. Model frameworks: most models are trained for Object Detection but lack the capability for producing more precise pixel-masks. This can limit accuracy and inference precision in the designed models. Additionally, many of those able to use pixel-masks are trained on low amounts of dataset images, limiting their accuracy as well.
2. Crowdmapping manual requirements: many applied crowdmapping systems using these models lack fully autonomous methods of uploading images, meaning individuals have to tag images according to their contents or at least approve of the generated model inferences.

The design process we implement looks to improve both issues as follows: by creating Instance Segmentation models for detection, we can improve accuracy which will then allow for more autonomous and reliable crowdmapping.

4 Engineering Design Goal

The goal of the project is to train and apply Computer Vision systems in Instance Segmentation for classifying and detecting local litter products based on image surveys of a given region. These models will then be used in allowing for crowdmapping of litter products. These systems enable automatic litter density mapping and will help aid in recognizing which regions contain the highest litter densities.

While this project is classified as environmental engineering and follows a design process rather than strict research question, the following is a hypothesis for the results of model training:

Hypothesis: If Instance Segmentation using one of the most state-of-the-art frameworks along with enough training dataset images and optimized hyperparameterization is used to train a set of CV models, then the result will achieve a level of accuracy usable for real-world litter mapping applications.

5 Methods

The core of our project takes place under the following design steps:

1. Compile a variety of publicly available annotated image datasets for model training
2. Assign training classes (training, validation and testing) at random to images
3. Augment and preprocess images
4. Assign training parameters
5. Train models
6. Analyze results and change parameters as well as dataset requirements as needed.

Below is an explanation of these steps.

In compiling the datasets, a variety of designated waste product datasets were used in addition to the public availability of the datasets in Roboflow Universe, a website for locating and collecting images for Computer Vision applications. The annotation type was polygon annotations, consisting of exact pixel-mask boundaries around each instance of an object class within an image. More than 25 individual datasets were included across the 7 compiled datasets for this task (exact notes on the sources used are available in the project notebook). Many significant external datasets (not accessed from Roboflow Universe) include the TACO dataset, BePLi v2, GSA2D-2024 and ZeroWaste datasets, each with high image counts and relevance for the applied object classes.

One emphasis on deciding on the used metrics was in looking for real-world images of litter, as these are more applicable and help to improve model robustness. All of the previously mentioned datasets help to achieve this goal, and provide a wide ranging variety of visual conditions for which the models are trained upon.

The Training, Validation and Testing split for models is a critical step in maximizing computational and dataset resources while being able to prioritize model safety (limiting incorrect predictions) or improving detection of the object class over prioritizing avoiding False Positives. Images used for training are those which the models are exposed to during training instead of for evaluating model accuracy. These are the main basis for building model weights and therefore require the largest portion of all images. The images for Validation are used throughout the model training to determine how effective the model is at detecting the object class. These images are therefore also important but not the primary training basis. Test images have the least value and are only used after training to check total progress. However, they do allow for more efficient and accurate comparisons. Considering the relative importance of each type of image classification, the split for 70:20:10 was decided between Training, Validation and Testing respectively in order to optimize the limited image dataset quantities that were available.

Preprocessing is a way of slightly adjusting images in the datasets in order to increase the ability of models to train properly with these images. As such, two preprocessing steps were used, these being auto-alignment of annotations and image resizing to 640 by 640 pixels. While auto-alignment is used to prevent errors in dataset image annotations, the resizing was a more crucial step in order to standardize the model input and therefore improve training throughout the process. Augmentations

were only used in selective object classes in order to artificially add images where certain classes lacked high image counts, meaning so they could have sufficient training data where a lack of publicly available datasets existed.

Model parameters are discussed in the next section based on prior research on the benefits of certain hyperparameters.

Model training was straightforward and completed through access to a remote GPU (namely the NVIDIA Tesla L4 GPU for this use) with Google Colab in order to improve training speed. All models were trained under the same hyperparameters as shown below in order to improve reproducibility.

Analyzing the results against current results is shown in section 7.

The parameters used during training are as follows (each individual parameter is explained in the next section):

1. Training for 325 epochs
2. Patience value of 50 epochs
3. Pretraining using yolov11n-seg.pt
4. Weight optimizer with AdamW
5. Initial learning rate of 0.001
6. Batch size 32, 8 workers, image size 640 pixels by 640 pixels

5.1 Justification of Model Hyperparameterization

The training period of 325 epochs implies that each model reviews each image in the Training and Validation sections of the datasets exactly 325 times in order to train weights. This period is a balance between overfitting where the model trains to specifically perform well on the dataset and underfitting in which the model does not fully understand the dataset before training concludes. Previous work within the YOLO framework has shown that between 250 and 350 epochs is a good duration training period to fully understand the training dataset and the upper range is used when the image count is high.

Patience of 50 epochs means that the model training would stop if it does not improve within a interval of the last 50 epochs in order to avoid wasting resources on training that does not improve model accuracy. This only applied to certain models, as shown in the Results Section.

The yolov11n-seg.pt pretrained model is the lightest of the YOLOv11 series in terms of accuracy but by far the fastest. In order to improve the real-world applications, it was chosen over certain other "heavier" models to increase model speed for these applications.

The weight optimizer of AdamW is one of the most commonly used optimizers for CV applications and was therefore employed in model training. It is commonly paired with a learning rate of 0.001 to allow for adjustments and rapid model progress as training progresses. In YOLO frameworks, models follow adapted model weights in training where weights contribute to the output of a model after training. For AdamW, weights are determined by the following formula:

$$\theta_t = (1 - n\lambda)\theta_{t-1} - n \left(\frac{m_t}{\sqrt{v_t} + \epsilon} \right)$$

where θ_t is the adapted weight, θ_{t-1} is the preceding weight, n is the learning rate at this stage of training, m_t and v_t are the bias-corrected first and second moment moving gradient averages respectively and ϵ is a small constant to prevent division by 0. The main benefits of AdamW are that it is adaptive, has decoupled weight decay and is computationally efficient. Compared to optimization functions like Adam which only employs L2 Regularization rather than this type of weight adjustment in this case, AdamW was the best choice for model optimization.

Batch size of 32 and 8 workers was chosen in order to optimize the efficiency of model training and use the high resources available for these models based on the Google Colab resources provided. These parameters determine how many CPU resources are dedicated at any time to model training. Maintaining relatively high values helps to improve the effective use of available resources.

Standardized image size is a commonly employed way of making inputs more similar such that the models are able to improve their progress more regularly during training. Such standardization

corresponds with the image dataset sizes and helps to improve accuracy during training. Additionally, square image sizes are particularly effective in promoting robust model detections and help to prevent input image size from being a limiting factor in accuracy.

5.2 Efficiency Metrics

Inference speed refers to the time which a model takes to fully analyze and apply its weights to an input and is an important metric for real-time model comparisons. The YOLO series of CV models has been long favored for its high inference speeds, especially with recent models such as YOLOv11. Some models are able to achieve more than 100 frames per second of detections when run on GPUs, making YOLO a good candidate for real-time CV applications. Even with these high speeds, making sure that model accuracy does not compromise speed is still a major concern, especially for cases when a GPU is unavailable. Because of the need for real-time applications, maintaining high inference speeds will be a goal throughout this development and used to validate model progress along with accuracy.

5.3 Accuracy Metrics

The metrics of Precision, Recall, mAP and F1 Scores were all used to determine the accuracy of trained models. The basis for all of these metrics is Intersection over Union (IoU) which measures the shared area of each of a model's predictions divided by the total area of both the prediction and ground truth, only counting the shared area once. Precision and Recall are measures of individual detections and are calculated at a certain IoU value. F1 Scores are the harmonic mean of Precision and Recall and help to determine the balance between False Negatives (FN) and False Positives (FP). The formula for each is as follows (TP is True Positive):

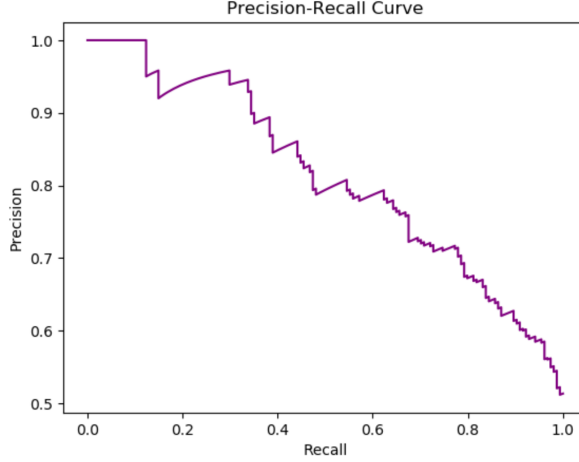
$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \times P \times R}{P + R}$$

mAP values measure how well a model performs at a certain IoU and confidence restriction. Confidence values are associated with model predictions and show how confident a model is for a certain prediction. For a set of IoU values, the Precision and Recall are calculated across many confidence restrictions which only consider confidence values above a certain value (confidence values are given by these systems along with the visual predictions and object class designations). The metric of mAP50 only refers to the IoU value of 50 while mAP50-95 is an average of all average Precision values from IoU restrictions of 0.5 to 0.95 at 0.05 intervals. The metric of mAP50-95 is generally more difficult to be at a higher value because as IoU restrictions increase, model predictions need to be more precise in order to be determined as accurate. Then, when these values are graphed, the area under the graph is determined as the mAP value. The graph itself is referred to as a Precision-Recall Curve from this process. A combination of all of these metrics is utilized to determine how each model performs comparatively to one another.

Below is an example of such a curve used in mAP calculations.



$$mAP = \int_0^1 (Precision(R)dR)$$

where R is a Recall value at a given confidence threshold. We can write Precision as a function of R as Recall values are taken continuously across all confidence thresholds and, because of restricting confidence values in considering valid predictions, the values of Precision and Recall are continuously defined by each other. For example, if we set a confidence value low, then Recall will be very high and Precision will be correspondingly low (as shown by largely increasing the total number of predictions). If we increase the confidence threshold, then the number of overall predictions will decline as a result. This leads to Recall decreasing but Precision increasing. As a result, the Precision function becomes continually defined for each distinct possible Recall value as shown through changing confidence thresholds.

6 Results

To begin this section, we have included the approximate number of images used per dataset in training, as well as the estimated number of specific objects in total across all images that were annotated:

Table 2: Dataset Counts

Object Class	Image Counts	Annotated Object Counts
Plastic Bottles	15000	31000
Drink Cans	8000	15000
Plastic Bags	7500	10000
Food Containers and Wrappers	5000	12500
Plastic Cups	8000	15000
Plastic Films	7500	12500
Drinking Straws	5000	9000

There is a total of 56000 images across all of these classes allowing for robust training for detection. Any quantity of images greater than 3000 to 4000 is considered an especially good basis for model accuracy, meaning 56000 images has great potential for applications particularly for real-world issues. The datasets were selected to prioritize a variety of backgrounds between real world and otherwise which helps to improve model prediction robustness. The emphasis on more than 1 object per image is a good basis to increase the detection of all instances of an object in an image as well.

As for the accuracy results, the metrics showed positive results compared to the baseline applications. Below is the current results for all training sessions where P is Precision values and R is Recall values:

Table 3: Accuracy metrics

Object Class	P	R	F1	mAP50	mAP50-95
Plastic Bottles	0.89	0.73	0.802	0.808	0.7
Drink Cans	0.962	0.806	0.877	0.866	0.815
Plastic Bags	0.863	0.542	0.62	0.619	0.516
Food Containers and Wrappers	0.871	0.733	0.8	0.819	0.741
Plastic Cups	0.946	0.905	0.925	0.953	0.888
Plastic Films	0.916	0.776	0.84	0.85	0.716
Drinking Straws	0.944	0.87	0.905	0.939	0.837

Additionally, below are the inference speeds when ran on a CPU (namely the Intel Core i7-9700 CPU) for each model:

Table 4: Accuracy metrics

Object Class	Inference Speed (ms)
Plastic Bottles	113
Drink Cans	105
Plastic Bags	66
Food Containers and Wrappers	72
Plastic Cups	66
Plastic Films	130
Drinking Straws	75

These values were obtained as the average detection time for all images within the testing datasets after training the models. The combined detection time on average for each image is therefore 627 milliseconds, meaning roughly 100 images can be fully analyzed each minute and 6000 can be analyzed in an hour. This standard is highly efficient for real-world detections and maintains the low inference times associated with the YOLO architecture.

Note below as well some of the produced inferences of the models (all base images were taken from the TACO dataset):



Figure 2: Plastic Bottle Model Predictions



Figure 3: Drink Can Model Predictions



Figure 4: Plastic Bag Model Predictions



Figure 5: Food Container and Wrapper Model Predictions



Figure 6: Plastic Cup Model Predictions



Figure 7: Plastic Film Model Predictions



Figure 8: Drinking Straw Model Predictions

7 Analysis

Inference speed is more variable depending on the computational resources. The use of a GPU can reduce inference speed significantly (in our case it could take as low as 0.3 milliseconds to perform inference for one model) compared to the use of a CPU. However, applications on devices with limited resources require speeds to be maintained along with accuracy, therefore implying that inference speed is still a significant factor for these applications. As a result, using a CPU, the models maintained a rough average of 100 milliseconds per image, meaning that together, it would take less than a second for detection of all available object classes within an image. This performance is highly acceptable and surpasses many of the current applications of similar systems for both accuracy and inference speed (as shown in section 7.1). This performance is likely highly acceptable for the desired applications of the system as a similar CPU will likely be available for using this software.

An additional consideration is the use of pixel-masks that these models are able to produce. By being able to precisely detect the boundaries of the objects in the images, the following benefits are presented:

1. Readily available annotations: if further polygon annotations were required for training more models, then these could be taken directly from the output of Instance Segmentation models, therefore meaning that for applications in tasks such as pretraining for further analysis or expanding datasets for manual detection, these models have stronger use.
2. More precise predictions: The exact objects are shown within the image instead of just regions that may contain these objects, meaning showing the exact boundaries prevents misinterpreted information about the images.
3. Beneficial object size information: Analysis based on the size of objects predicted in determining predictions accuracy is allowed as a result of being able to access the exact pixel-masks to show the area or relative size of objects. This can be developed into a filter for predictions using object size conditions to further increase accuracy regarding certain object types.

Now consider the following aspects of the analysis:

7.1 Comparison to Models in Literature Review

The below table is complete with the available accuracy metrics across all works mentioned in the Literature Review in order to determine the relative accuracy of our models:

Table 5: Accuracy metrics

Author	Object Class	P	R	F1	mAP50	mAP50-95
Ours	Plastic Bottles	0.89	0.73	0.802	0.808	0.7
Ours	Drink Cans	0.962	0.806	0.877	0.866	0.815
Ours	Plastic Bags	0.863	0.542	0.62	0.619	0.516
Ours	Food Containers and Wrappers	0.871	0.733	0.8	0.819	0.741
Ours	Plastic Cups	0.946	0.905	0.925	0.953	0.888
Ours	Plastic Films	0.916	0.776	0.84	0.85	0.716
Ours	Drinking Straws	0.944	0.87	0.905	0.939	0.837
(Mandhana, N 2026)	Drink Cans	x	x	x	0.3351	0.177
(Mandhana, N 2026)	Plastic Films	x	x	x	0.214	0.177
(Heller, A 2025)	Plastic Bottles	0.843	0.538	0.657	0.611	0.444
(Jaikumar, P 2022)	Plastic Bottles	x	x	x	0.761	0.6
(Akkoyun, D 2024)	Drink Cans	0.74	0.63	0.68	0.64	x
(Akkoyun, D 2024)	Plastic Bags	0.62	0.4	0.49	0.766	x
(Akkoyun, D 2024)	Plastic Cups	0.68	0.59	0.63	0.64	x
(Akkoyun, D 2024)	Drinking Straws	0.96	0.64	0.768	0.766	x
(Akkoyun, D 2024)	Plastic Films	0.72	0.7	0.71	0.64	x
(Dipo, M 2025)	Plastic Bottles	0.707	0.707	0.707	0.78	0.65
(Shad, I 2025)	Food Container and Wrapper	0.77	0.68	0.72	0.785	0.654

Note how for (Akkoyun, D 2024), because of how models were compiled, mAP metrics were calculated across all classes at the same time in two sets, resulting in the metrics for the Drink Cans, Plastic Cups and Plastic Films along with Plastic Bags and Drinking Straws models respectively being equal. The same applies to (Mandhana, N 2026) except only for mAP50-95. In any instance of an "x" in the table, this metric was not completed for the given paper, due to differing framework standards.

An analysis by each object class goes as follows:

7.1.1 Plastic Bottles

This is a historically heavily represented litter class across models, thus implying strong prior developments to this object class' detection systems. The three comparisons included with (Heller, A 2025), (Jaikumar, P 2022) and (Dipo, M 2025) show that our system has strong improvements upon the baseline for this subfield. Our F1 scores of 0.802 and mAP50 scores of 0.808 surpass the current standards in the range of 0.657 to 0.707 currently across the three previous papers. The relatively low Recall value on our model corresponds to the variability of quality of these objects in real-world settings, as shown by the inability to track all instances of these object with a score of 0.73. While this may be comparatively low to other models in our framework for other object classes, this is typi-

cally acceptable for real-world applications especially with both the high Precision value and normally greater image quality for such applications.

7.1.2 Drink Cans

As another heavily represented class, our drink can model had varying comparative metrics against the two other instances of similar models. Its mAP values vastly exceeded those presented in (Mandhana, N 2026) as this paper relied on a smaller dataset for training, and our model had higher overall metrics compared to (Akkoyun, D 2024) despite a closer difference compared to other object classes.

7.1.3 Plastic Bags

Our model performed worse in mAP50 compared to the benchmark (Akkoyun, D 2024), but had highly exceeding Precision, Recall and F1 scores compared to the previous work. However, the large gap between Precision and Recall indicates that more training data may be required in order to further improve the model to maintain a level of accuracy similar to other object class models that were developed in this design process. However, because Precision values which determine False Positives are still at an acceptable range of 0.863 in this object class, it is unlikely to interfere with the combined model predictions along with the other 6 models. This is because the failure to detect a plastic bag as represented by Recall does not produce inaccurate object detections by themselves, it only fails to recognize an object. Therefore, very few mislabeled litter products will result from this model when used together in the system.

7.1.4 Food Containers and Wrappers

We maintained narrowly higher metrics with a very similar mAP value compared to the previous work. Despite our slight edge over the previous model framework, increasing the size of the training dataset could improve accuracy to a higher degree, as indicated by the somewhat low Recall score.

7.1.5 Plastic Cups

This model outperformed the previous model, with higher metrics across those available. This object class has historically been overshadowed through either only plastic bottle detection or mixed models capable of detecting both types of products. Therefore, providing a specialized model for this purpose is an added benefit of this engineering process as the distinction between plastic bottles and plastic cups is significant in terms of purpose and how both items are most commonly purchased as well as littered into the environment.

7.1.6 Plastic Films

This was among the more underrepresented classes in CV literature and only contained few previously designed models that functioned well. While (Mandhana, N 2026) relied on fewer training images and therefore maintained lower mAP metrics, our model metrics when compared against (Akkoyun, D 2024) show improvement from the current benchmark, with a roughly 0.2 increase in Precision, 0.17 increase in Recall, 0.195 increase in F1 scores and 0.299 increase in mAP50.

7.1.7 Drinking Straws

This was additionally one of the more underrepresented classes, especially regarding dataset availability, which therefore made model training difficult without sourcing images from a wider variety of datasets as a result. However, when compared to (Akkoyun, D 2024), our model had a slightly lower Precision value but overall still surpassed the current model basis within this object class with a much higher F1 score and mAP values.

7.1.8 General Analysis across all Models

More generally across all object classes, our models performed either at a comparable level or exceeded the benchmarks across these models. Additionally, many of the previous models were trained for Object Detection instead of Instance Segmentation, meaning that these higher metrics show a positive use

of polygon annotations and this specific framework compared to the more commonly used Object Detection frameworks. This supports a benefit of the different model training variation and supports our hypothesis (provided in section 4) regarding accuracy differences between models.

Regarding the models that could benefit from more training, the object class of plastic bags was an outlier for both previous literature and our designed model, in that its accuracy was significantly lower than other models for similar tasks. This reveals a weakness in dataset availability as well as an increased challenge with this detection type, this being the appearance of this object which could be more difficult for standard CNN-based CV models to detect. Therefore, later progressions of this project should focus within this object class for improving accuracy.

7.2 Model Performance for Waste Detection Applications

When considering real-world applications, the most valuable metrics are Precision and Recall (and therefore by extension F1 scores) such that models are able to detect the presence of objects in the image. Metrics like mAP, being based on IoU, are generally less important for litter detection comparisons as detecting the presence of objects is more valuable than knowing how well the exact detections are within the images for such objects. As a result, through strictly noting F1 scores, the developed models show promising results along with the efficiency metrics shown in Section 6 for applications in litter detection systems. Across all models, these values were at least 0.8 for all but 1 (the outlying plastic bag model that was developed additionally) with some achieving extremely high relative scores of 0.905 and 0.925 which show a high level of accuracy needed in these real-world applications.

7.3 Evaluation of Instance Segmentation over Object Detection

As mentioned previously, the existing frameworks with Object Detection were generally outperformed by this use of Instance Segmentation which, generally promotes this as an effective method for improving model accuracy besides increasing dataset size or changing hyperparameters. Especially for non-rectangular object classes such as plastic bags, drinking straws or food wrappers, due to polygon annotations having a higher level of precision, the results show that this method could help with detecting irregular object shapes as a result of more accurate object boundaries.

8 Conclusions and Continuing Work

The accuracy and relative efficiency of the proposed model frameworks across 7 object classes shows a high level of performance for real-world applications. Besides this, it serves as evidence for the improved usage of Instance Segmentation systems over the historically dominant Object Detection systems that are more typical in CV applications for similar systems.

In reference to the main contributions of our work, we accomplished the following:

1. A fully autonomous detection system capable of recognizing the vast majority of littered plastic products in real-world image data.
2. A design for crowdmapping that reduces manual effort and allows for large regions of litter to be classified more efficiently.
3. Evidence supporting Instance Segmentation being a more accurate method of CV for litter detection (and likely many other object classes) when compared to Object Detection.

8.1 Current Crowdmapping Applications

We are using the designed system for an effective crowdmapping integration as follows:

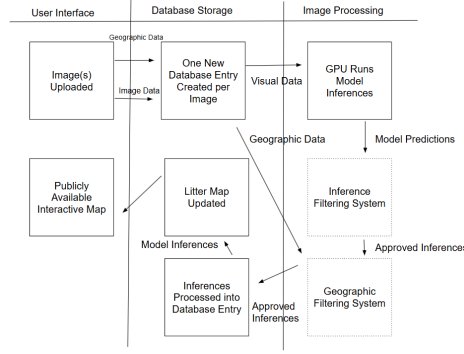


Figure 9: CV Automated Crowdmapping Data Flow Design

Both of the dashed surrounding boxes are unique additions to crowdmapping with limited prior developments. The design functions on the basis of user input as directed through a web interface. This interface built in the form of a web application, will allow for users to upload images taken of litter from their local environments. These images must be accompanied with geographic metadata as well as when the images were taken, both accessible directly from the image without needing separate files for it to function. This is uploaded to a database where these images are stored, with a single entry being a single image. From there, the images have inference run on them by the set of 7 Instance Segmentation models, and these inferences are filtered based on overlapping class identifications and confidence rates. Then, the accepted inferences enter a system of filtering based on the location and time in which they were taken. For example, an inference from an image taken within a 1 meter radius of the same item with a relatively recent time frame from the previous image (for example, within 1 day), would be removed under this system to prevent duplicates. Once all inferences are processed and filtered, the finalized image annotations are uploaded back to the database and the litter map is updated on a regular basis (within 1 to 2 minutes) to ensure accurate predictions. The users are then able to view the images in a way such that they can find densities and use this information for coordinated efforts at removing litter or determining which regions have the highest densities.

While similar applications do exist, below are the proposed benefits of this design over the current standards:

1. Complete automation: some major applications such as OpenLitterMap employ mostly manual image tagging in order to ensure images are properly annotated. Our design relies upon automated CV models which have demonstrated robust levels of accuracy (above 0.8 F1 and 0.8 mAP50 metrics for most) implying increased convenience and reduced time per upload for users. Additionally, for standard crowdmapping without image upload, there is no way to confirm detections as well as determine whether or not the detected objects are unique and have not been previously uploaded. This system improves upon both completely manual and semi-manual detection systems of crowdmapping as a result.
2. An effective framework for accuracy-efficiency balance: our models are capable of achieving superior accuracy compared to the benchmarks within 1 second of inference time. This framework is highly effective for potential usage in low-computational resource environments and expands the access to such systems.
3. A proposed application structure with model filtering: by using Instance Segmentation models, having more precise inference information allows for visual area filters between model predictions that could potentially be used to further accuracy across models in a way that Object Detection models are unable to. For example, in our model filtering system, we will use both standard information about overlapping model predictions as well as how large predictions are to estimate revised confidence values in producing outputs. Additionally, the geographic filtering system is a needed proposal for crowdmapping that utilizes both when and where images were taken to reduce duplicates in model predictions. This will likely increase accuracy above what the model metrics point to directly from the results.

4. Multi-object tagging: Because of the high efficiency, simply having someone locate a region of high litter densities and take a more general image instead of having to classify each object they see actually reduces the time that crowdmapping takes. Further optimizations with autonomous systems for taking visual data can also provide a completely autonomous system in this case.

8.2 Web Application Overview

The following is an early implementation of a web application that could be used for crowdmapping:

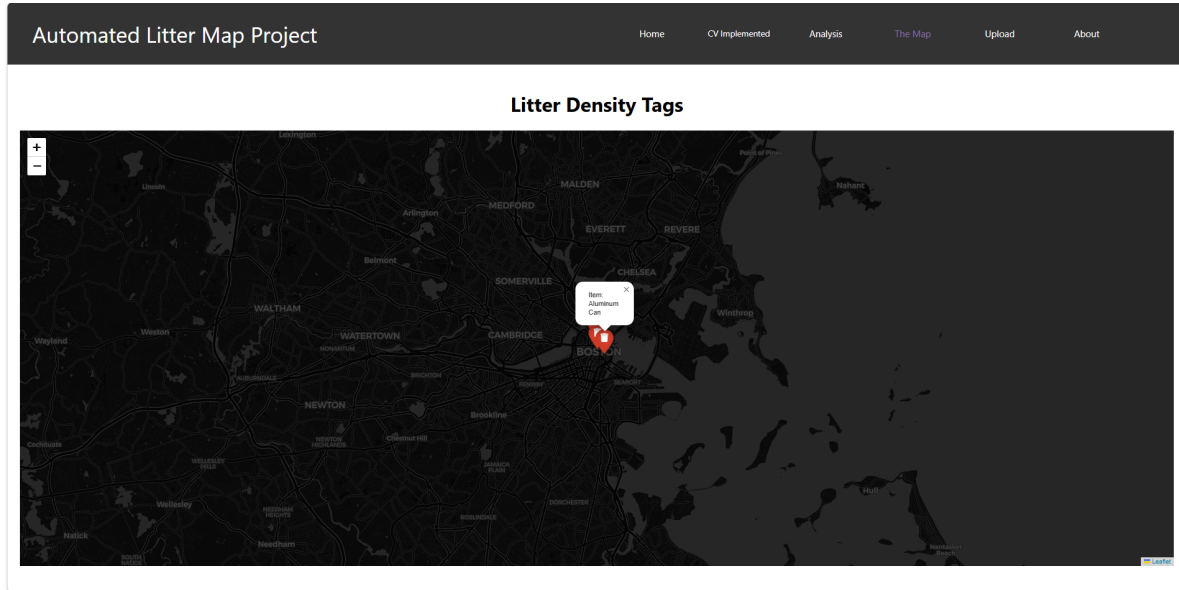


Figure 10: Website for Crowdmapping Early Implementation

The website consists of 6 tabs, of which the sections titled "The Map" and "Upload" will be used for an interactive overlay for users. Images uploaded to the webpage will be stored in a database and go through the process as shown in Figure 9 in order to have all litter products detected. Then, the individual markers for these products will appear in an interactive map that users can view.

8.3 Limitations and Future Work

Although many of the waste products that this project analyzed are commonly seen in urban areas, with only 7 object classes it becomes difficult to be able to detect all sources of litter, even if these 7 products contain a much more significant portion of all litter that exists in many regions. As a result, further applications or work within this project may focus on increasing its scope to products such as cigarettes or other common litter products for urban regions, with the goal of being able to automatically identify all of these products at an effective speed accurately.

An additional limitation was the lack of available polygon annotated data for these image classes. Despite the high number of used images, improved accuracy could still come with more openly available real-world projects focused in this goal. Many further applications of this project may however help this issue, as being able to recycle annotated images through model predictions is a powerful tool for the generation of future datasets for similar object classes.

The model accuracy, while highly acceptable for this application, could still be improved through having access to higher computational resources. However, without these resources, it becomes difficult to manage high accuracy while maintaining low enough training times. The design product is still highly accurate and surpasses several of the current benchmark standards for this type of detection, meaning future applications of this system are likely to be very accessible and improve upon the current work for crowdmapping.

While the applications of these systems are currently limited to crowdmapping, other applications could involve using UAV data for detections in a wider region, thus increasing the inputted visual data

for which models are able to analyze. However, such systems require intense planning in terms of UAV restrictions and navigation, and will therefore likely be employed after optimizing the crowdmapping web application for which these models were initially designed.

Beyond litter detection, the models also show promise for waste sorting applications at a potentially higher rate than the more chaotic outdoor environments in which these systems are to be used. The automation of sorting collected trash would benefit significantly from this software especially due to its nature as Instance Segmentation which allows for more precise inference for uses such as physically sorting items out of conveyor belts. As a result, if such an opportunity is available, application in such a regard may be pursued, but remains separate from the goal of mapping litter densities that this project focuses on.

Acknowledgments

I submitted this work to the MSEF as a representative of North Reading High School’s World of STEM student organization. Additional support came from North Reading’s local Green Alliance group and North Reading High School’s EcoTeam in navigating this topic. The Adult Sponsor for this project was one of the World of STEM group’s teacher supervisors. The author greatly appreciates these groups’ support throughout the project.

No external mentor heavily guided the work done for this project (barring the inspiration and supervision of the Adult Sponsor and other club members) and it was completed independently. No external funding or university affiliation was involved in the project.

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